

Extra exercises of Lecture 21

EX 1

Both bank X and Y have the same level of liabilities valued at \$6,9 M, but bank X and Y total assets are valued at \$7.5 M and at \$7.7 M (respectively), and their capital is \$600k and \$800k (respectively). Which bank is riskier if risk-weighted assets are \$2,5 M for both banks?

- a) It cannot be determined
- b) Bank X is as risky as bank Y
- c) Bank X is riskier than bank Y
- d) Bank Y is riskier than bank X

EX 2

Bank A has the following assets: reserves of \$100, treasury bonds of \$200, residential mortgages of \$400, and corporate loans of \$50. Bank A's net worth is \$100. Bank B has the following assets: reserves of \$50, treasury bonds of \$400, residential mortgages of \$100, municipal bonds of \$400 and corporate loans of \$50. Bank B's net worth is \$120. Assume the following risk weights apply:

Reserves and High-quality government bonds	0%,
High-quality government bonds and Residential mortgages	0%
Residential mortgages and Municipal bonds	50%
Corporate loans	100%

Which bank will be considered riskier according to the Basel methodology?

- a) both banks have the same level of risk
- b) the relative riskiness of the two banks cannot be determined from the information provided
- c) bank B is riskier
- d) bank A is riskier

EX 3

A bank has deposits for which it pays an interest equal to $i_d > 0$ and it has some positive capital ($NW > 0$). The bank can invest in one of the two projects. A safe project A with a positive return $R_A > i_d$ and a project B with an expected positive return $E(R_B)$ such that with a given probability $0 < p_B < 1$ it yields R_B , and 0 otherwise. Assume that deposits are fully insured by the government. If the return on project A increases of 20%, what happens to the capital requirement that makes the bank indifferent between investing in project A or B?

- a) it increases
- b) it decreases
- c) it does not change
- d) it is not predictable with the given data

EX 4

Bank New-Lending assets are \$30 cash, \$70 Municipal bonds and \$100 loans, with risk weights respectively of 0, 0.5 and 1. The bank has liabilities of \$185. Regulatory authorities have imposed two requirements: 1st) a ratio of bank capital to liabilities of at least 9%, and 2nd) a total capital requirement of 8% of risk weighted assets.

Which of the two requirements Bank New-Lending does comply with?

- a) requirements 1st) and 2nd)
- b) neither requirement
- c) requirement 1st) but not 2nd)
- d) requirement 2nd) but not 1st)